

4.13.99

AI25BTECH11030 -Sarvesh Tamgade

Question: Two planes are given:

$$P_1 : 10x + 15y + 12z - 60 = 0, \quad (0.1)$$

$$P_2 : -2x + 5y + 4z - 20 = 0. \quad (0.2)$$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ?

$$(a) \frac{x-1}{0} = \frac{y-1}{-5} = \frac{z-1}{4}$$

$$(b) \frac{x-6}{-5} = \frac{y-2}{3} = \frac{z}{-5}$$

$$(c) \frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$$

$$(d) \frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$$

Solution:

The two planes are:

$$P_1 : \mathbf{n}_1^\top \mathbf{r} = d_1, \quad \mathbf{n}_1 = \begin{pmatrix} 10 \\ 15 \\ 12 \end{pmatrix}, \quad d_1 = 60, \quad (4.1)$$

$$P_2 : \mathbf{n}_2^\top \mathbf{r} = d_2, \quad \mathbf{n}_2 = \begin{pmatrix} -2 \\ 5 \\ 4 \end{pmatrix}, \quad d_2 = 20. \quad (4.2)$$

A general line is

$$\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d}. \quad (4.3)$$

For a line to be an edge of a tetrahedron whose two faces lie on P_1 and P_2 , it must satisfy one of the conditions:

$$t_1 = \frac{d_1 - \mathbf{n}_1^\top \mathbf{r}_0}{\mathbf{n}_1^\top \mathbf{d}}, \quad t_2 = \frac{d_2 - \mathbf{n}_2^\top \mathbf{r}_0}{\mathbf{n}_2^\top \mathbf{d}}, \quad (4.4)$$

and then either

- 1) $\mathbf{n}_1^\top \mathbf{r}_0 = d_1$ and $\mathbf{n}_1^\top \mathbf{d} = 0$ i.e. the line lies on plane P_1 , or
- 2) $\mathbf{n}_2^\top \mathbf{r}_0 = d_2$ and $\mathbf{n}_2^\top \mathbf{d} = 0$ i.e. the line lies on plane P_2 , or
- 3) $t_1 \neq t_2$ i.e. the line intersects both planes on two different points.

We test the options:

$$(a) \mathbf{r}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} 0 \\ -5 \\ 4 \end{pmatrix}. \mathbf{n}_1^\top \mathbf{r}_0 = 37, \mathbf{n}_2^\top \mathbf{r}_0 = 7, \mathbf{n}_1^\top \mathbf{d} = -27, \mathbf{n}_2^\top \mathbf{d} = -9,$$

$$t_1 = \frac{60-37}{-27} = -\frac{23}{27}, t_2 = \frac{20-7}{-9} = -\frac{13}{9}, \quad t_1 \neq t_2 \text{ (valid).}$$

$$(b) \mathbf{r}_0 = \begin{pmatrix} 6 \\ 2 \\ 0 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} -5 \\ 3 \\ -5 \end{pmatrix}. \mathbf{n}_1^\top \mathbf{r}_0 = 90, \mathbf{n}_2^\top \mathbf{r}_0 = -2, \mathbf{n}_1^\top \mathbf{d} = -65, \mathbf{n}_2^\top \mathbf{d} = 5,$$

$$t_1 = \frac{60-90}{-65} = \frac{6}{13}, t_2 = \frac{20-(-2)}{5} = \frac{22}{5}, \quad t_1 \neq t_2 \text{ (valid).}$$

$$(c) \mathbf{r}_0 = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} -2 \\ 5 \\ 4 \end{pmatrix}. \mathbf{n}_1^\top \mathbf{r}_0 = 60, \mathbf{n}_2^\top \mathbf{r}_0 = 20, \mathbf{n}_1^\top \mathbf{d} = 103, \mathbf{n}_2^\top \mathbf{d} = 45,$$

$$t_1 = \frac{60-60}{103} = 0, t_2 = \frac{20-20}{45} = 0, \quad t_1 = t_2 \text{ (invalid).}$$

$$(d) \mathbf{r}_0 = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}. \mathbf{n}_2^\top \mathbf{r}_0 = 20, \mathbf{n}_2^\top \mathbf{d} = 0,$$

$$(-2 \ 5 \ 4)\mathbf{r} = 20 \quad \forall t, \quad (\text{the line lies on } P_2) \text{ valid.}$$

Therefore, the valid edges are **(A)**, **(B)**, and **(D)**.

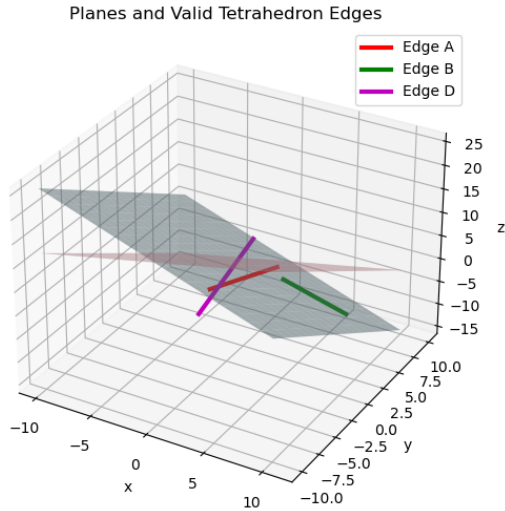


Fig. 4.1: Vector Representation